Group Project Info
CASA0029 Urban Data Visualisation

1. Project Output Table

Project Output	Output Description
Project Output Files	Zip file submitted on Moodle
Project Website	https://hhyoloo.github.io/CASA0029-group-project/
Project Methodology Summary	The project methodology is summarised in the Word/PDFfile submitted with the Project Output Files.

2. Individual Contributions Table

Task Name	Major Contributors	Additional Contributors	Use of AI Tools in this Task
Concept Development and overall project planning	All team members		No AI use
Indicator design (BHP), Interactive Visualisation 3 and Project Methodology	Xuchen Xi		ChatGPT was used only for limited language refinement during drafting; methodological design and coding decisions were independently developed and checked against literature
Interactive Visualisation 1, statistical charts and website text writing	Xiaoyi Wang		AI tools were used only to improve wording clarity in explanatory text; chart design and visual outputs were manually produced and adjusted
Interactive Visualisation 2 and website framework	Yutong Xu		ChatGPT was used to assist front-end debugging, layout refinement and interface adjustment; final structure and interaction logic were manually revised
Website content integration	Xiaoyi Wang, Xuchen Xi	Yutong Xu	ChatGPT was used selectively for wording refinement only

Urban Flyways in London

1. Research Background and Urban Analytics Theme

Following the CASA0029 Urban Analytics theme, this project explores the ecological rhythms of London through seasonal bird migration. In this context, the city is not only seen as a human-centred built environment, but also as part of a wider ecological system that supports migratory species at different times of the year. Urbanisation is often associated with biodiversity loss (Marzluff, 2001), so this study looks at how urban landscapes may influence the spatial distribution and clustering of three representative species: the Common Swift, Barn Swallow, and House Martin. The core research question is: how can urban data analytics reveal spatiotemporal patterns in bird migration, and how might these patterns relate to urban environmental conditions? The project moves from temporal patterns to spatial hotspots, with the aim of providing a basis for interpretation in an urban planning context.

2. Data Sources and Spatial Processing

The project combines bird observation data with a set of urban environmental variables. Biological data was drawn from large-scale observation records, from which three species were selected to capture variation in behaviour. Global analyses indicate that urbanisation exerts a significant filtering effect on bird diversity (Aronson et al., 2014). Environmental variables include building density, access to larger green spaces, and vegetation cover (NDVI). These were selected because they represent both ecological support and urban pressure. Green infrastructure is vital for linking landscapes and communities (Benedict and McMahon, 2006). Spatial processing was carried out using GeoPandas to ensure efficient rendering during web-based interaction.

3. Indicator Modelling & MCDA

The main methodological component of the project is the Bird Habitat Pressure (BHP) index, developed using a Multi-Criteria Decision Analysis (MCDA) approach (Malczewski, 2006). The aim of this indicator is to combine different environmental factors into a single layer that can support interpretation.

3.1 Data Normalization

As the input variables have different units, Min–Max normalisation was applied to rescale all values to a common range between 0 and 1. This allows the variables to be combined without one dominating due to scale differences.

$$X_{norm} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

3.2 BHP Comprehensive Evaluation Model

The BHP index combines indicators of ecological support with indicators of urban pressure, producing a relative measure of habitat conditions across space. In general terms, areas with higher ecological quality and lower disturbance are expected to show lower habitat pressure, while the opposite indicates higher pressure. The composite score is calculated using the following formula:

$$BHP = (w_1 \cdot H_i + w_2 \cdot C_i) - (w_3 \cdot D_i + w_4 \cdot L_i)$$

An equal weighting scheme is used when combining variables to keep the method transparent, as there is no clear basis for assigning species-specific weights at this scale. However, this also represents a simplification, as the relationship between environmental conditions and bird behaviour is likely to be more complex. The index should therefore be interpreted as an indicative measure rather than a precise ecological model.

Key Variable Definitions and Theoretical Support:

Habitat Quality (H_i): Represents local habitat support, including greener or more ecologically favourable conditions. Green infrastructure serves as a foundational asset for urban biodiversity (Benedict and McMahon, 2006).

Connectivity (C_i): Captures whether the area can function as part of a wider ecological movement network. Connectivity is a vital element for maintaining biological flows within a landscape structure (Taylor et al., 1993).

Human Disturbance (D_i): Approximates anthropogenic load such as dense built fabric and direct urban disturbance. Anthropogenic landscapes are a primary driver behind the loss of urban birds (Marzluff, 2001).

Light Pollution (L_i): Represents artificial night-time light exposure that can disturb orientation, feeding, and nesting behaviour. Bright lights in big cities pose severe exposure risks to migratory birds (Horton et al., 2020) and impact animal physiology broadly (Dominoni et al., 2020).

4. Visual Narrative and Interactive Logic

The website is structured to guide users through different stages of interpretation. It begins with temporal patterns, allowing users to see how bird activity changes across seasons, then moves to spatial hotspot mapping. Users can then explore environmental layers through the "Habitat Lens", comparing hotspots with underlying urban conditions. Finally, the BHP index groups areas into categories—Urban Pressure Area, Attention Zone, Potential Corridor, and Stable Habitat Core—translating data into an interpretable form linked to planning discussions.

5. Collaboration & AI Ethics Statement

The project was developed collaboratively, with different team members contributing to data preparation, visualisation, and interface design. AI tools were used in a limited way to assist with coding and wording, but the overall methodology, indicator design, and interpretation were developed by the group.

References

【1】 Marzluff, J. M. (2001). Urbanization and the loss of birds. In: Avian Ecology and Conservation in an Urbanizing World.

【2】 Dominoni, D. M., et al. (2020). Why light and noise pollution matter for animals. Biological Reviews.

【3】 Horton, K. G., et al. (2020). Bright lights in the big cities: migratory birds' exposure to artificial light. Nature Communications.

【4】 Taylor, P. D., et al. (1993). Connectivity is a vital element of landscape structure. Oikos.

【5】Benedict, M. A., & McMahon, E. T. (2006). Green Infrastructure: Linking Landscapes and Communities.

【6】Malczewski, J. (2006). GIS-based multicriteria decision analysis: a survey of the literature. International Journal of Geographical Information Science.

【7】Aronson, M. F. J., et al. (2014). A global analysis of the impacts of urbanization on bird and plant diversity. Proceedings of the Royal Society B.